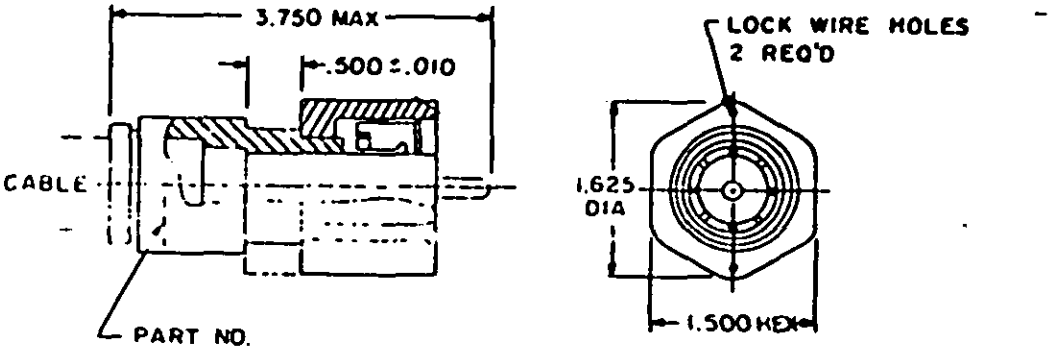
MILITARY SPECIFICATION SHEET

CONNECTOR, PLUG, ELECTRICAL, SERIES LT, CABLED, CLASS T

[NOT FOR SPACECRAFT APPLICATIONS]

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cables described herein shall consist of this specification and the latest issue of MIL-C-25537(USAF).

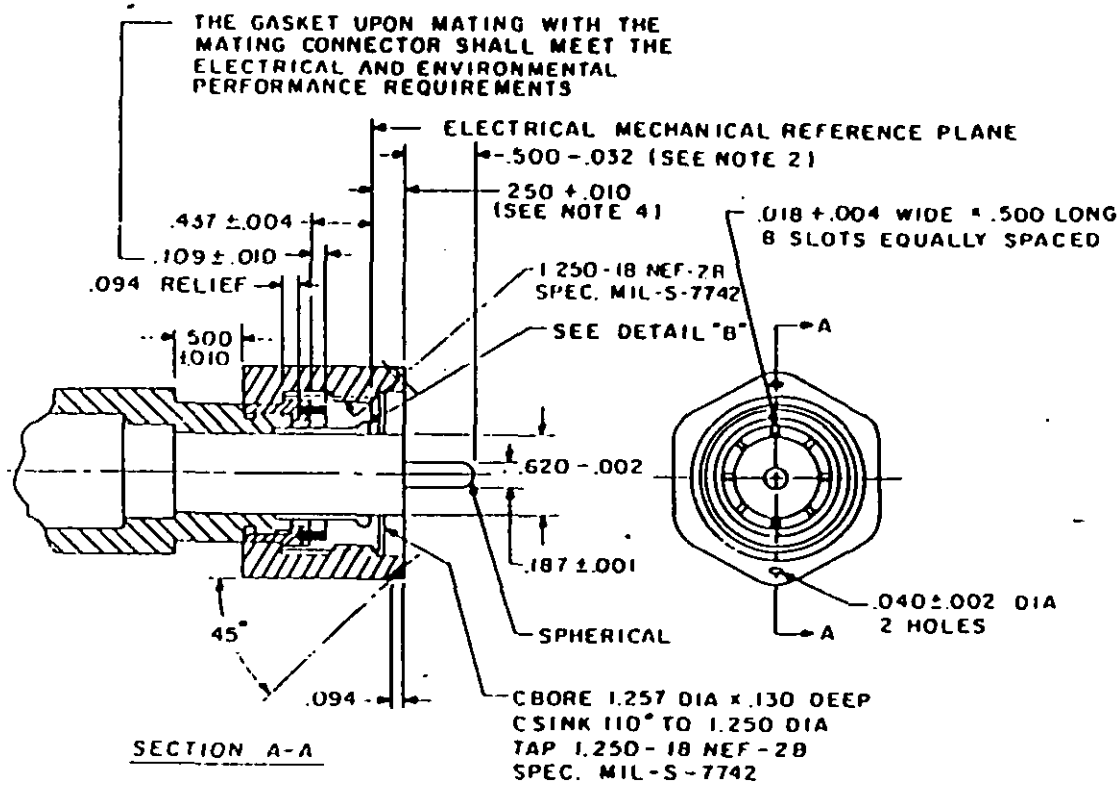


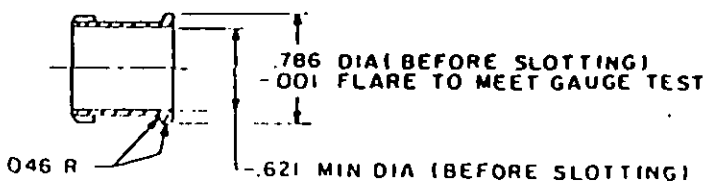
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension 1.625 (41.28 mm) is the largest overall diameter of the connector.
4. Wrench flats, when applicable, are to accommodate standard wrench opening in accordance with FSC-STD-H28.
5. All undimensioned pictorial representations are for reference purposes only.
6. Dimension 3.750 (95.25) defines the maximum length of the connector when assembled to the appropriate cable.

FIGURE 1. General configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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DETAIL "B"

NOTES:
1. Dimensions are in inches.
2. This dimension is from the tip of the center contact to the end of the dielectric.
3. Unless otherwise specified, all tolerances shall be ±.005 inch.
4. This dimension is from the end of the outer contact to the end of the dielectric.

FIGURE 2. Mating dimensions for pin terminations.

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Connector durability: 500 cycles minimum at 12 cycles per minute maximum. The connector shall meet the mating characteristics and force to engage and disengage requirements.

Contact resistance (in milliohms maximum):

	Initial	After environment
Center contact:	1.5	2.0
Outer contact:	.2	Not applicable
Braid to body:	2.0	Not applicable

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 2,500 volts rms minimum at sea level.

Vibration, high frequency: Method 204 of MIL-STD-202, test condition B. No discontinuity permitted.

Shock: Method 213 of MIL-STD-202, test condition I. No discontinuity permitted.

Thermal shock: Method 107 of MIL-STD-202, test condition B, except test high temperature shall be +85°C. High temperature shall be +200°C for connectors using +200°C cables.

Moisture resistance: Method 106 of MIL-STD-202. No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:

Altitude: 70,000 feet.

Cable	V rms
M17/72-RG211 M17/161-00001	1700

RF high potential withstanding voltage: (1100 V rms after moisture vibration and shock).

Frequency: 5 to 7.5 MHz inclusive.

Leakage current: Not applicable.

Cable	V rms
M17/72-RG211 M17/161-00001	10000, at sea level 2500, at 70,000 feet

Cable retention:

Cable	Force
M17/72-RG211 M17/161-00001	210 pounds

Power handling capability: 500 watts CW at 5 GHz (max. temperature rise not to exceed 20°F when tested at 70,000 feet at 203°F).

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ENGINEERING DATA:

Nominal impedance: 50 ohms.

Frequency range: 0 to 5 GHz.

Voltage rating:

500 volts rms maximum working voltage at sea level.

125 volts rms maximum at 70,000 feet.

Temperature rating: -65°C to +105°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 6 inch-pounds maximum.

Coupling proof torque: 15 inch-pounds minimum.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 6 to 10 inch-pounds.

Mating characteristics: See figure 2 for dimensions.

Outer contact:

Test ring I.D.: .786 maximum, 16 microinches finish.

Insertion force: 25 pounds maximum when inserted a minimum of .093.

Contacts with slotted members: Shall contact a .794 minimum diameter ring within .031 of their tip ends and shall not contact a .804 minimum inside diameter ring.

Insulation resistance: Method 302 of MIL-STD-202, test condition B, 5,000 megohms minimum.

Corrosion (salt spray): Method 101 of MIL-STD-202, test condition B.

Voltage standing wave ratio (VSWR): From 0.5 to 5.0 GHz, 1.15 +.030 x frequency (GHz).

Swept frequency VSWR test setup:

Item 6: VSWR shall be less than 1.02 from dc to 1 GHz and less than 1.05 from 1 to 5 GHz.

Item 16: VSWR shall be less than 1.02 from dc to 1 GHz and less than 1.05 from 1 to 5 GHz.

Second step of VSWR checkout procedure. VSWR shall be less than 1.10 + .006 F (F in GHz).

Group B inspection: VSWR shall be less than 1.10 + .012 F (F in GHz).

First article inspection: VSWR shall not exceed 1.15 + .030 F (F in GHz).

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Coupling mechanism retention force: 100 pounds minimum.

Part number: M26637/1-1.

TABLE I. Supersession data.

Preferred part number	Superseded part number
M26637/1-1	MS24281-1

Custodian:
Air Force - 85

Review activity:
Air Force - 99

User activity:
Air Force - 11

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-F281-011)

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